

CHEMURNAUT, Russian Federation

WMO: 259410

Lat: 60.733N

Lon: 163.017E

Elev: 46

StdP: 14.67

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.9	-22.5	-32.5	0.9	-26.8	-28.0	1.2	-22.2	41.4	18.6	37.4	4.5	13.6	160	0.924

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	65.8	57.5	62.9	56.2	60.6	54.6	59.1	64.3	57.1	61.8	55.4	59.6	11.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	67.7	62.0	54.4	63.2	60.2	52.7	59.3	58.4	25.8	64.7	24.5	61.9	23.5	59.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
35.9	31.9	28.9	-30.2	72.9	4.5	5.4	-33.4	76.8	-36.0	80.0	-38.6	83.1	-41.8	87.0		
			-29.8	63.4	4.3	4.4	-32.9	66.6	-35.4	69.1	-37.8	71.6	-40.9	74.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	2.4	4.4	7.1	21.8	35.0	47.4	53.9	53.6	45.7	31.3	17.8	5.6
	DBStd	21.18	12.67	13.65	12.07	8.17	4.98	5.01	4.14	3.50	5.05	8.51	12.26	11.89
	HDD50	8581	1476	1276	1330	847	466	107	8	9	140	579	967	1376
	HDD65	13767	1941	1696	1795	1297	931	528	346	353	578	1044	1417	1841
	CDD50	292	0	0	0	0	1	29	128	122	12	0	0	0
	CDD65	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH74	7	0	0	0	0	0	0	7	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	12.8	14.8	14.6	14.0	13.3	11.2	11.3	10.7	10.8	11.0	12.9	14.5	14.5
Precipitation	PrecAvg	25.6	1.9	1.8	1.9	1.3	1.2	1.6	2.0	3.0	2.6	2.9	3.1	2.3
	PrecMax	32.2	4.0	3.3	4.0	3.2	2.8	3.5	4.1	5.2	5.2	5.0	7.7	5.2
	PrecMin	18.9	0.2	0.4	0.4	0.3	0.4	0.6	0.4	0.8	0.9	1.3	1.0	0.7
	PrecStd	3.5	0.9	0.8	0.9	0.6	0.6	0.8	0.9	1.3	1.1	0.8	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.1	36.6	31.9	37.3	53.7	67.6	74.1	68.7	59.5	48.6	39.4	35.1
		MCWB	33.1	34.9	30.3	33.3	44.1	55.2	64.7	59.8	54.0	44.5	37.1	33.2
	2%	DB	33.2	31.5	29.8	35.2	47.0	62.7	66.8	64.9	56.9	46.3	37.3	30.3
		MCWB	31.4	29.8	28.4	32.5	41.4	52.9	58.7	57.9	52.1	43.0	35.2	28.7
	5%	DB	27.1	28.0	27.0	33.9	44.4	59.7	63.5	62.1	54.9	44.4	35.6	26.3
		MCWB	25.7	26.5	25.5	31.8	39.5	51.1	57.0	56.8	50.7	41.6	33.6	24.6
	10%	DB	19.5	23.2	23.7	32.3	42.1	56.8	61.1	59.8	53.0	42.6	33.4	22.1
		MCWB	18.1	21.9	22.2	30.3	38.2	49.4	55.3	55.1	49.4	40.2	31.6	20.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	34.8	30.6	34.2	45.5	58.0	65.0	61.4	54.9	45.4	37.5	33.5
		MCDB	34.7	36.2	31.6	35.9	52.5	66.4	72.8	66.5	58.5	47.6	39.3	34.9
	2%	WB	31.3	29.9	28.6	33.1	41.8	53.9	59.8	58.9	52.8	43.2	35.4	28.9
		MCDB	33.2	31.4	30.0	34.6	46.4	60.8	65.4	63.5	56.3	45.7	36.9	30.1
	5%	WB	25.8	26.6	25.4	31.9	40.0	51.7	57.7	57.1	51.1	41.8	33.7	24.6
		MCDB	27.2	28.2	27.0	33.8	43.9	58.9	62.5	61.3	54.3	44.2	35.4	26.2
	10%	WB	18.2	21.9	22.2	30.4	38.4	49.9	55.8	55.5	49.6	40.2	31.7	20.5
		MCDB	19.5	23.2	23.6	32.3	41.8	55.8	60.3	59.4	52.6	42.5	33.5	22.1
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.4	9.9	8.3	8.3	12.8	11.2	10.1	9.6	8.4	9.6	9.2
		MCDBR	9.7	11.3	9.5	7.2	12.6	19.8	17.2	15.2	12.9	8.3	9.5	12.2
		MCWBR	9.0	10.8	9.1	6.0	8.9	13.2	11.7	10.9	10.0	7.4	9.0	11.4
	5% WB	MCDBR	9.4	11.2	9.3	6.8	11.5	18.5	15.7	13.9	11.0	7.6	9.2	11.9
MCWBR		9.3	10.8	9.0	5.7	8.4	13.0	11.7	10.6	9.3	7.1	9.0	11.5	
Clear-Sky Solar Irradiance	taub	0.226	0.272	0.332	0.398	0.449	0.411	0.446	0.432	0.343	0.317	0.260	0.209	
	taud	2.052	2.106	2.008	1.890	1.936	2.243	2.174	2.189	2.443	2.289	2.093	1.959	
	Ebn at Noon	165	224	242	246	243	259	245	237	246	206	144	114	
	Edh at Noon	13	23	37	50	52	39	41	38	24	20	13	9	
All-Sky Solar Radiation	RadAvg	112	368	867	1429	1671	1656	1558	1125	743	350	117	52	
	RadStd	16	27	85	138	162	188	163	107	61	42	16	5	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page